Sampadram Kumar Jogadenu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.60
Narayana Junior College, 122823A41, Gottipatitowers, 4THFL, Santhoshnagar, Mehdipatnam	Class XII (Telangana State Board of Intermediate Education)	2023	98.90%
Narayana School Kukatpally, Medchal District	Class X (Board of Secondary Education, Telangana)	2021	10.00%

POSITIONS OF RESPONSIBILITY

Sunshine, IIT Hyderabad - Member of Newsletter team

Teaching Assistant:

- Worked under Dr. Anjana Mahesh as a Teaching Assistant for the Probability and Random Variables course.
- Conducted tutorial sessions, graded assignments and examinations, and assisted students with course-related problems

SKILLS

Programming skills: C, C++, Python, SQL, LaTeX

Libraries: Numpy, Matplotlib, PyTorch

Git, GitHub

RELEVANT COURSES

AI: Computer Vision, NLP, Deep Learning, Online Learning, Reinforcement Learning

CS: Computer Architecture, DBMS, Data Structures and Application, Algorithms, Software Engineering

Mathematics: Optimization, Probability and Random Variables, Applied Statistics, Matrix Theory (Linear Algebra)

PROJECTS

Indian Handicrafts Portal - NLP Course Project, Link

- Built a Next.js handicraft discovery portal with searchable, filterable, and state-based exploration across 200+ Indian handicrafts.
- Developed a similarity-based duplicate detection workflow and prototype NLP capabilities using TF-IDF search, MiniLM embeddings, and extractive text summarization.
- Created an interactive chatbot, India craft map, detailed craft pages, and community registration workflow with browser-based persistence.

Seq2Seq Transliteration (*Dataset : Dakshina - Telugu Lexicons)*

- Implemented a seq2seq model using encoder-decoders with additive attention
- Implemented batching to accelerate training and support variable batch sizes.
- Applied teacher forcing during training to enhance convergence and improve sequence learning
- Achieved an exact word accuracy of approximately 60% for the validation and test set using the above configuration

Deep Reinforcement Learning, Link

- Implemented a CNN-based DQN agent for Atari Pong using grayscale 84×84 frames and four-frame stacking.
- Built experience replay, epsilon-greedy exploration, target-network learning, Huber loss, Adam optimization, and gradient clipping in PyTorch.
- Added checkpoint-based training resumption and learning-curve evaluation, with the agent reaching an approximately +5.5 mean reward over a rolling 50-episode window.

Satellite Image Semantic Segmentation Using Multispectral U-Net, Link

- Built a PyTorch U-Net for multispectral semantic segmentation using 16-channel input imagery.
- Implemented image preprocessing, channel conversion, normalization, and validation splitting for training.
- Trained with Dice loss and mixed precision, using checkpointing to retain the best-performing model.

Test-Time Poisoning Attack on ViTs, Link

- Studied the cross-architecture transferability of test-time poisoning attacks from CNN surrogate models to ResNet-50 and Vision Transformer models.
- Implemented and evaluated SAR and ROID test-time adaptation under sequential ImageNet-C distribution shifts, comparing clean and adversarial performance.
- Developed a grey-box poisoning attack using entropy sharpening, cross-entropy optimization, and feature-statistics drift to degrade model accuracy under adversarial streams.

Analytics BI, Backend Developer - Software Engineering Course Project, Link

- Contributed to Analytics BI, a full-stack business intelligence platform for ingesting, cleaning, querying, visualizing, and dashboarding CSV/Excel data.
- Designed and implemented the Data Transformation Services pipeline to clean, normalize, and semantically validate ingested records, including type, nullability, date, range, and constraint validation.
- Built the Dead Letter Queue and quarantine workflow to capture invalid records with structured error details, categories, severity, fingerprints, suggested fixes, and row-level restore/validation operations.
- Implemented asynchronous background ingestion using BullMQ, streaming/batched processing, worker threads, and Socket.IO progress events for large CSV/XLS/XLSX uploads.

EXTRACURRICULAR

Volunteering: National Service Scheme (NSS), IIT Hyderabad - 140 Hours

Hobbies: Table tennis, Video gaming, Digital Art

EXPERIENCE

Project Intern, Oracle

- Developed a **metadata-grounded NL-to-SQL system** for querying enterprise **PeopleSoft database** using natural-language questions.
- Implemented **Graph Traversal and Hybrid Retrieval** approaches for schema grounding, combining lexical, semantic, attribute, and entity-based retrieval.
- Built a **governed SQL generation pipeline** with schema allowlists, trusted joins, read-only validation, and deterministic SQL rendering to prevent unsafe or invalid queries.
- Developed an **AI-assisted result reporting layer** that summarizes query results and presents business-oriented explanations of the returned data.